Class No: 03

Control Flow – Conditions & Loops

Conditional Statements in JavaScript:

Conditional statements in JavaScript are used to make **decisions based on conditions**. If a condition is true, a specific block of code is **executed**; otherwise, another block of code runs.

If-Else Statement

This is the most basic conditional statement. If a condition is true, one block of code will execute; otherwise, another block will execute.

Example:

```
let age = 18;

if (age >= 18) {
    console.log("You are an adult.");
} else {
    console.log("You are a minor.");
}
```

Else-If Ladder

The else if statement allows you to test multiple conditions.

Example:

```
let marks = 85;
if (marks >= 90) {
    console.log("Grade: A+");
} else if (marks >= 75) {
    console.log("Grade: A");
} else if (marks >= 60) {
    console.log("Grade: B");
} else {
    console.log("Grade: C");
}
```

Ternary Operator

The ternary operator is a shorthand way of writing an if-else statement.

Syntax:

```
condition ? expression_if_true : expression_if_false
```

Example:

```
let age = 18;
let result = (age >= 18) ? "Adult" : "Minor";
console.log(result);                    // Adult
```

More Conditions

- If
- Ifelse
- elseif
- Switch
- Check Positive or Negative Number, 2.Check if Number is Zero
- Check Even or Odd, 4.Check Passing Marks
- Check Voting Eligibility, 6.Check Age for Driving
- Find Greater Number, 8.Smallest of Two Numbers
- Check if Number is Divisible by 5, 10. Check if Number is Divisible by 10
- Check if Character is Capital Letter,12. Check if Temperature is Hot or Cold
- Check if Person is Teenager 14. check the student fail or pass

Loops in JavaScript

What are Loops?

Loops allow you to repeat a block of code multiple times until a specific condition is met.

1. For Loop

A for loop repeats a block of code a fixed number of times.

Syntax:

```
for (initialization; condition; increment) {
        // Code to execute
}
```

Example:

```
for (let i = 0; i < 5; i++) {
    console.log(i);                // Prints 0 to 4
}
```

While Loop

A while loop repeats a block of code as long as the condition is true.

Syntax:

```
while (condition) {
    // Code to execute
}
```

Example:

```
let i = 1;
while (i < 5) {
    console.log(i);                    // Prints 0 to 4
    i++;
}
```

Do-While Loop

A do-while loop is similar to a while loop, except that it executes the block at least once, even if the condition is false.

Syntax:

```
do {
    // Code to execute
} while (condition);
```

Example:

```
let i = 0;   0
do {
    console.log(i);                    // Prints 0 to 4
    i++;
} while (i < 5);
```

Key Differences Between Loops

Loop Type	Description	Example Use Case
For Loop	Repeats a block a fixed number of times.	Iterating over an array.
While Loop	Repeats a block as long as a condition is true.	Validating user input.
Do-While Loop	Executes the block at least once.	Asking for input until it's valid.

continue and break keyword

- <u>Continue</u>
- It **skips the current iteration** of the loop.
- The loop will **continue with the next iteration.**
- <u>Break</u>
- It **completely stops the loop.**
- The loop will **not run anymore** after break.
- Return
- Used **inside a function.**
- It **stops the function immediately** and sends a value (or nothing) back.

Key Differences:

Feature	break	return
Where used?	Loops & switch	Functions
Effect	Stops only the **loop/switch**	Stops the **entire function**
Can send value?	❌ No	✅ Yes, returns a value
Execution after	Continues after the loop	Function exits immediately

Activity-Based Task

Activity: Number Guessing Game

Task Description:

We will create a simple **Number Guessing Game** in class.

- The program will generate a random number between 1 and 10.
- Students will use a loop to guess the number until they get it right.

Code for the Activity:

```
let randomNumber = Math.floor(Math.random() * 10) + 1;
let guess;

do {
    guess = Number(prompt("Guess a number between 1 and 10:"));
    if (guess > randomNumber) {
        console.log("Too high! Try again.");
    } else if (guess < randomNumber) {
        console.log("Too low! Try again.");
    }
} while (guess !== randomNumber);

console.log("Congratulations! You guessed the correct number: " + randomNumber);
```

Home Task 3

- What is a conditional statement?
- Write a program to check if a number is positive, negative, or zero.
- What is the difference between if-else and ternary operator?
- Write a script to print all even numbers from 1 to 10 using a for loop.
- Write a JavaScript program to check if a number is divisible by 5 using a while loop.
- Explain the difference between while and do-while loops with examples.
- Write a program that prints the multiplication table of a number using a for loop.
- Create a program to count the number of digits in a given number using a while loop.
- Write a program to print numbers from 10 to 1 using a for loop (reverse order).
- Write a script that asks the user to input numbers until the sum of all entered numbers is greater than 100.

Thank You